

Project Final Report

SmartSDLC – AI-Enhanced Software Development Lifecycle

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Tools Used: VS Code, Streamlit, FastAPI, Hugging Face API, Python, IBM Granite model

1. Abstract

SmartSDLC is an AI-powered assistant that automates and enhances key stages of the Software Development Lifecycle. It enables users to upload PDF documents or input textual prompts to extract software requirements, generate clean backend/frontend code in various frameworks, and produce corresponding test cases. This tool leverages Large Language Models (LLMs) to accelerate and streamline early-stage development.

2. Problem Statement

Software development often begins with time-consuming requirement analysis, manual code drafting, and writing of test cases. These tasks, though crucial, slow down productivity and increase the chance of human error. There is a need for an intelligent solution that simplifies and automates these processes.

3. Objectives

- To simplify and automate software requirement analysis from PDF or text input
 - To generate backend or frontend code using AI in user-selected languages/frameworks
 - To automatically generate unit and functional test cases
 - To provide a clean and intuitive interface for developers
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4. Features

-  **PDF/Prompt-Based Requirement Analysis**
Upload a PDF or type a project idea to extract key functional and non-functional requirements.
-  **AI Code Generation**
Automatically generate code by specifying the language (e.g., Python, JS) and framework (e.g., Flask, React).
-  **Test Case Generator**
Produce relevant unit and functional test cases based on code or requirement context.

5. Architecture Diagram

csharp

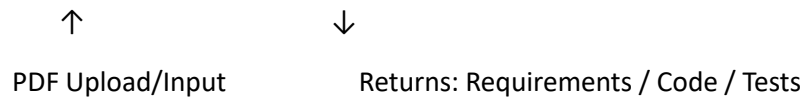
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[Replace this section with your system architecture diagram or use the below description]

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User → Streamlit Web Interface → Backend (FastAPI) → Logic Layer (Python) → Hugging Face API (IBM Granite Model)



6. Modules

1. **PDF/Prompt Parser** – Parses and extracts content from uploaded PDFs or text.
2. **Requirement Extractor** – Uses IBM Granite LLM to identify functional & non-functional requirements.
3. **Code Generator** – Generates code using LLM with selected language and framework.
4. **Test Case Generator** – Outputs unit/functional test cases from given code or description.
5. **Web UI** – Built with Streamlit for easy user interaction.

7. Technologies Used

- Python
- Streamlit
- FastAPI
- Hugging Face Transformers
- IBM Granite (LLM)
- PyPDF2
- VS Code

8. Sample Screens

Add screenshots of your Streamlit app interface, code generation output, test case generation, etc.

9. Conclusion

SmartSDLC enables developers to reduce effort in repetitive early SDLC stages using AI. By combining Hugging Face's LLM capabilities with a user-friendly frontend, the tool delivers requirement analysis, code generation, and test creation — all in one platform. It demonstrates how AI can transform and accelerate modern software development.